

Real_Estate_Data

Table of contents

Libraries	1
Load Map Data	3
Load Dataset	3
Average Land Prices	3
Hsinchu	3
Taipei City	5
Elderly Dependency Ratio and Average Land Prices	6
Hsinchu	6
Bar Plot of Average Land Prices with Elderly Dependency Ratio as Second Scale	7
Hsinchu Linear Regression Model	8
Scatter Plot	9
Taipei City	10
Bar Plot of Average Land Prices with Elderly Dependency Ratio as Second Scale	11
Taipei Linear Regression Model	12
Scatter Plot	13

Libraries

```
library(sf)
```

Linking to GEOS 3.13.1, GDAL 3.11.0, PROJ 9.6.0; sf_use_s2() is TRUE

```
library(dplyr)
```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

filter, lag

The following objects are masked from 'package:base':

intersect, setdiff, setequal, union

```
library(tidyr)
library(scales)
```

```
suppressMessages(suppressWarnings(
  source(here::here("R", "load_data.R"))
))
```

```
save_map_plot <- function(plot,
                           filename,
                           width = 13,
                           height = 8,
                           dpi = 300,
                           folder = "plots") {

  if (!dir.exists(folder)) {
    dir.create(folder, recursive = TRUE)
  }

  filepath <- file.path(folder, filename)

  ggsave(filepath,
    plot = plot,
    width = width,
    height = height,
    dpi = dpi)
}
```

Load Map Data

```
# Load township-level shapefile
map_data <- map_data[["TOWN_MOI_1140318.shp"]]
```

Load Dataset

```
# Look at the Metadata
manifest_list <- real_estate_data[["manifest.csv"]]
```

Average Land Prices

Hsinchu

In this section, we will try to analyze the elderly population data from Hsinchu City and County and see its relationship with the average land prices.

```
hsinchu <- list("j_lvr_land_a.csv", "o_lvr_land_a.csv") %>%
  lapply(function(file) {
    real_estate_data[[file]] %>%
      {
        colnames(.) <- as.character(unlist(.[,1, ]))
        .[-1, ]
      } %>%
    select(1, 13, 23)
  }) %>%
  bind_rows()

hsinchu <- hsinchu %>%
  filter(`main use` %in% c(" ", " ", " ")) %>%
  rename(TOWNNAME = `The villages and towns urban district`) %>%
  rename(LANDPRICE = `the unit price (NTD / square meter)`)

hsinchu_ave_price <- hsinchu %>%
  group_by(TOWNNAME) %>%
```

```

summarise(
  avg_price = mean(
    as.numeric(str_replace_all(LANDPRICE, ",", "")),
    na.rm = TRUE
  )
)

price_to_copy <- hsinchu_ave_price %>%
  filter(TOWNNAME == " ") %>%
  pull(avg_price)

hsinchu_ave_price <- hsinchu_ave_price %>%
  add_row(
    TOWNNAME = c(" ", " ", " "),
    avg_price = price_to_copy
  ) %>%
  filter(TOWNNAME != " ")

```

```

hsinchu_map_data <- map_data %>%
  filter(COUNTYNAME %in% c(" ", " ")) %>%
  select(3, 4, 8)

```

```

hsinchu_map_data <- hsinchu_map_data %>%
  left_join(hsinchu_ave_price, by = "TOWNNAME") %>%
  replace_na(list(avg_price = NA))

```

```

# hsinchu_map <- ggplot(hsinchu_map_data) +
#   geom_sf(aes(fill = avg_price), color = "white") +
#   geom_sf_text(aes(label = TOWNNAME), size = 3, color = "black") +
#   scale_fill_gradient(low = "lightblue", high = "darkred", na.value = "gray")
#   labs(
#     title = "Hsinchu Land Prices",
#     fill = "Avg Price (NTD)"
#   ) +
#   theme_minimal()
#
# hsinchu_map
# save_map_plot(hsinchu_map, "hsinchu_map_large.png")

```

Taipei City

```
taipei <- ("a_lvr_land_a.csv") %>%
  lapply(function(file) {
    real_estate_data[[file]] %>%
      {
        colnames(.) <- as.character(unlist(.[1, ]))
        .[-1, ]
      } %>%
    select(1, 13, 23)
  }) %>%
  bind_rows()

taipei <- taipei %>%
  filter(`main use` %in% c(" ", " ", " ")) %>%
  rename(TOWNNAME = `The villages and towns urban district`) %>%
  rename(LANDPRICE = `the unit price (NTD / square meter)`)

taipei_ave_price <- taipei %>%
  group_by(TOWNNAME) %>%
  summarise(
    avg_price = mean(
      as.numeric(str_replace_all(LANDPRICE, ",", "")),
      na.rm = TRUE
    )
  )
```

```
taipei_map_data <- map_data %>%
  filter(COUNTYNAME %in% c(" ")) %>%
  select(3, 4, 8)
```

```
taipei_map_data <- taipei_map_data %>%
  left_join(taipei_ave_price, by = "TOWNNAME") %>%
  replace_na(list(avg_price = NA))
```

```
# ggplot(taipei_ave_price, aes(x = reorder(TOWNNAME, avg_price), y = avg_price)) +
#   geom_col(fill = "steelblue") +
#   geom_text(aes(label = round(avg_price)),
#             hjust = -0.1,
```

```

#           size = 3.5) +
#   coord_flip() +
#   scale_y_continuous(labels = label_number(accuracy = 1)) +
#   labs(
#     x = "Town",
#     y = "Average Unit Price (NTD / square meter)",
#     title = "Average Unit Price by Taipei District"
#   ) +
#   theme_minimal() +
#   theme(plot.title = element_text(face = "bold"))

# taipei_map <- ggplot(taipei_map_data) +
#   geom_sf(aes(fill = avg_price), color = "white") +
#   geom_sf_text(aes(label = TOWNNAME), size = 3, color = "black") +
#   scale_fill_gradient(low = "lightblue", high = "darkred", na.value = "gray90")
#   labs(
#     title = "Taipei District Land Prices",
#     fill = "Avg Price (NTD)"
#   ) +
#   theme_minimal()
#
# taipei_map
# save_map_plot(taipei_map, "taipei_map_large.png")

```

Elderly Dependency Ratio and Average Land Prices

Hsinchu

```

hsinchu_pop_data <- aging_data %>%
  filter(COUNTY %in% c(" ", " ")) %>%
  select(2, 4, 10) %>%
  mutate(A65UP_A15A64_RAT = as.integer(A65UP_A15A64_RAT)) %>%
  rename(TOWNNAME = TOWN)

hsinchu_pop_data

```

```

# A tibble: 16 x 3
  COUNTY TOWNNAME A65UP_A15A64_RAT
  <chr>   <chr>         <int>

```

1	13
2	23
3	31
4	34
5	20
6	19
7	28
8	38
9	36
10	25
11	47
12	15
13	21
14	21
15	24
16	21

```
hsinchu_map_data <- hsinchu_map_data %>%
  left_join(hsinchu_pop_data, by = "TOWNNAME")

hsinchu_map_data$avg_price <- as.numeric(hsinchu_map_data$avg_price)
hsinchu_map_data$A65UP_A15A64_RAT <- as.numeric(hsinchu_map_data$A65UP_A15A64_RAT)
```

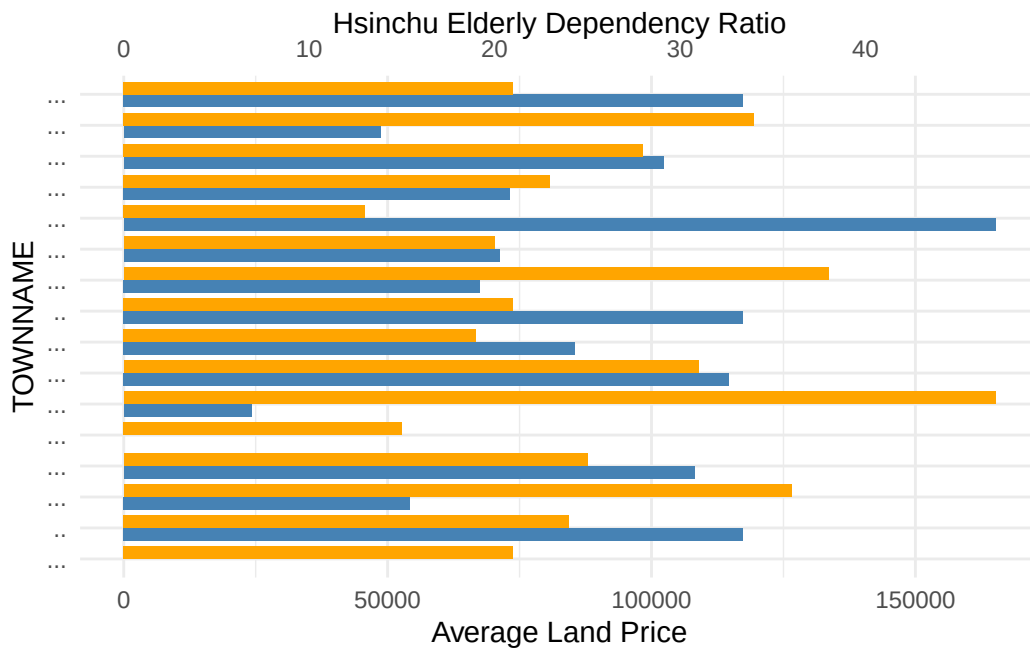
Bar Plot of Average Land Prices with Elderly Dependency Ratio as Second Scale

This plot demonstrates the comparison of the elderly dependency ratio vs average land prices per townships and districts in Hsinchu. This gives us an illustration of their relationship.

```
# Compute scaling factor
scale_factor <- max(hsinchu_map_data$avg_price, na.rm = TRUE) / max(hsinchu_map_data$A65UP_A15A64_RAT, na.rm = TRUE)

ggplot(hsinchu_map_data, aes(x = TOWNNAME)) +
  geom_col(aes(y = avg_price), fill = "steelblue", width = 0.4, position = position_nudge(x = 1)) +
  geom_col(aes(y = A65UP_A15A64_RAT * scale_factor), fill = "orange", width = 0.4, position = position_nudge(x = 2)) +
  scale_y_continuous(
    name = "Average Land Price",
    sec.axis = sec_axis(~./scale_factor, name = "Hsinchu Elderly Dependency Ratio")
  ) +
  coord_flip() +
  theme_minimal()
```

Warning: Removed 2 rows containing missing values or values outside the scale range (`geom\_col()`).



Hsinchu Linear Regression Model

```
hsinchu_model <- lm(avg_price ~ A65UP_A15A64_RAT, data = hsinchu_map_data)
summary(hsinchu_model)
```

Call:

```
lm(formula = avg_price ~ A65UP_A15A64_RAT, data = hsinchu_map_data)
```

Residuals:

Min	1Q	Median	3Q	Max
-41320	-17706	7896	13613	36076

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	174305.0	21248.1	8.203	2.91e-06	***
A65UP_A15A64_RAT	-3088.1	744.9	-4.146	0.00136	**

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 24450 on 12 degrees of freedom

(2 observations deleted due to missingness)

Multiple R-squared: 0.5889, Adjusted R-squared: 0.5546

F-statistic: 17.19 on 1 and 12 DF, p-value: 0.001357

Scatter Plot

The scatter plot shows us a better picture of the trend between the relationship of the elderly dependency ratio and the average land prices. In this case, we can see that the increase of the average land prices as an inverse relationship to the elderly dependency ratio.

```
ggplot(hsinchu_map_data, aes(A65UP_A15A64_RAT, avg_price)) +  
  geom_point(color = "steelblue", size = 3) +  
  geom_smooth(method = "lm", color = "orange", se = FALSE) +  
  labs(  
    title = "Elderly Dependency Ratio and Average Land Price in Hsinchu",  
    x = "Dependency Ratio",  
    y = "Average Land Price (NTD / sqm)"  
  ) +  
  theme_minimal()
```

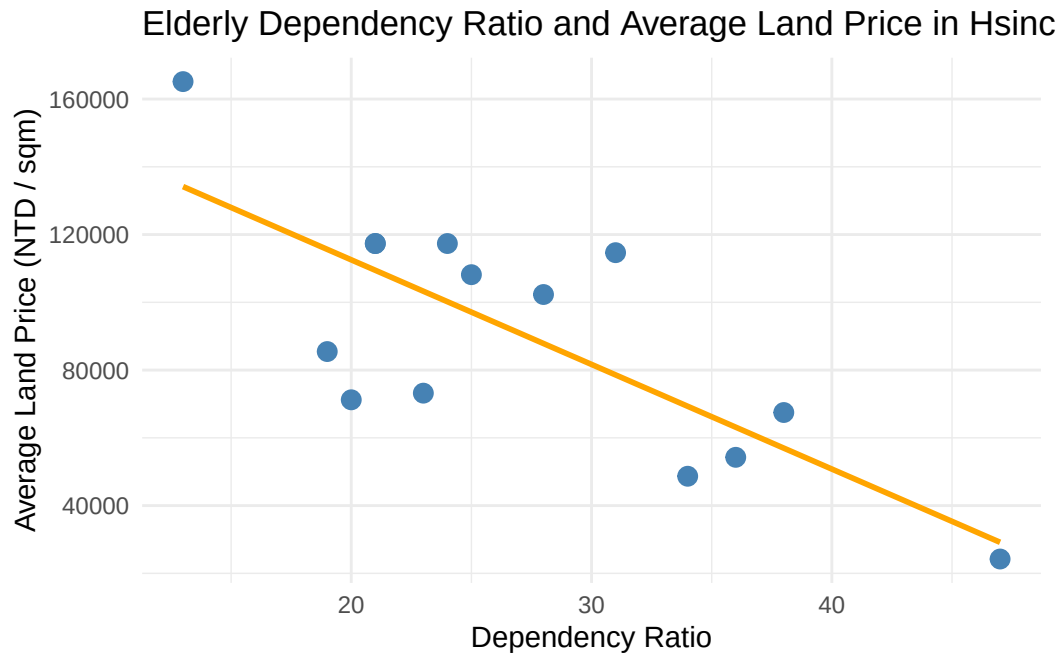
`geom\_smooth()` using formula = 'y ~ x'

Warning: Removed 2 rows containing non-finite outside the scale range

(`stat\_smooth()`).

Warning: Removed 2 rows containing missing values or values outside the scale range

(`geom\_point()`).



Taipei City

We will do the same process to Taipei City

```

taipei_pop_data <- aging_data %>%
  filter(COUNTY %in% c(" ")) %>%
  select(2, 4, 10) %>%
  mutate(A65UP_A15A64_RAT = as.integer(A65UP_A15A64_RAT)) %>%
  rename(TOWNNAME = TOWN)

```

taipei_pop_data

```

# A tibble: 12 x 3
  COUNTY TOWNNAME A65UP_A15A64_RAT
  <chr>   <chr>         <int>
1              38
2              38
3              41
4              37
5              36
6              35
7              38

```

8	31
9	31
10	28
11	37
12	33

```

taipei_map_data <- taipei_map_data %>%
  left_join(taipei_pop_data, by = "TOWNNAME")

taipei_map_data$avg_price <- as.numeric(taipei_map_data$avg_price)
taipei_map_data$A65UP_A15A64_RAT <- as.numeric(taipei_map_data$A65UP_A15A64_RAT)

```

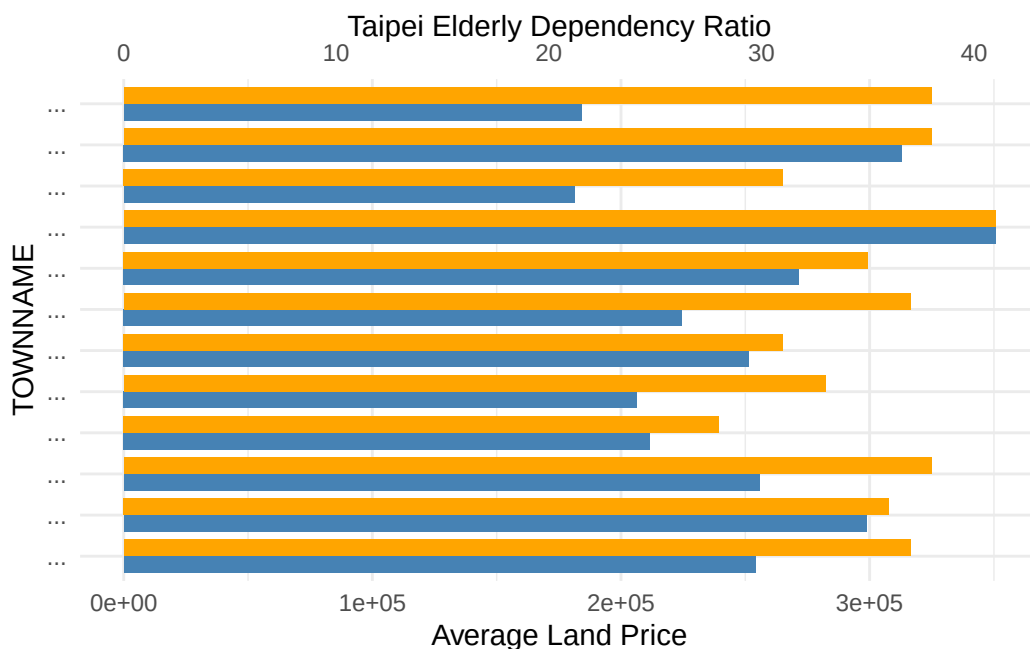
Bar Plot of Average Land Prices with Elderly Dependency Ratio as Second Scale

```

# Compute scaling factor
scale_factor <- max(taipei_map_data$avg_price, na.rm = TRUE) / max(taipei_map_data$A65UP_A15A64_RAT, na.rm = TRUE)

ggplot(taipei_map_data, aes(x = TOWNNAME)) +
  geom_col(aes(y = avg_price), fill = "steelblue", width = 0.4, position = position_nudge(x = 0.5)) +
  geom_col(aes(y = A65UP_A15A64_RAT * scale_factor), fill = "orange", width = 0.4, position = position_nudge(x = 1.5)) +
  scale_y_continuous(
    name = "Average Land Price",
    sec.axis = sec_axis(~./scale_factor, name = "Taipei Elderly Dependency Ratio")
  ) +
  coord_flip() +
  theme_minimal()

```



Taipei Linear Regression Model

```

taipei_model <- lm(avg_price ~ A65UP_A15A64_RAT, data = taipei_map_data)
summary(taipei_model)

```

Call:

```
lm(formula = avg_price ~ A65UP_A15A64_RAT, data = taipei_map_data)
```

Residuals:

Min	1Q	Median	3Q	Max
-87926	-28244	4524	36499	54637

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-29753	127358	-0.234	0.8200
A65UP_A15A64_RAT	7943	3594	2.210	0.0516 .

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 44930 on 10 degrees of freedom

Multiple R-squared: 0.3282, Adjusted R-squared: 0.261
F-statistic: 4.884 on 1 and 10 DF, p-value: 0.05155

Scatter Plot

In this scatter plot, we can see an interesting trend where the increase of land prices has a directly proportional to the elderly dependency ratio.

```
ggplot(taipei_map_data, aes(A65UP_A15A64_RAT, avg_price)) +  
  geom_point(color = "steelblue", size = 3) +  
  geom_smooth(method = "lm", color = "orange", se = FALSE) +  
  labs(  
    title = "Elderly Dependency Ratio and Average Land Price in Taipei",  
    x = "Dependency Ratio",  
    y = "Average Land Price (NTD / sqm)"  
  ) +  
  theme_minimal()
```

`geom\_smooth()` using formula = 'y ~ x'

